# ICT2132 · Object Oriented Programming Practicum

## Revision Guide and Practice

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

# Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

# 1. Main idea

Review separation of concerns, dependency injection, exception handling, JDBC parameter binding, transaction boundaries, and Swing event-thread behavior. Track what each test actually proves.

# 2. Example and reasoning

Practice: trace a button click to its SQL statement; write a mock service test; explain why a unit test is not a database persistence test; identify how a failed second insert should roll back the first.

# 3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

# 4. Questions for study

- Explain “revision guide and practice” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

## Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.